Lecture 24

- Stokes' theorem (16.8)

Recall: If $\vec{F} = (P, Q, R)$, then:

$$\operatorname{div}(\vec{F}) = \nabla \cdot \vec{F} = P_x + Q_y + R_z$$

$$\operatorname{curl}(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$$

$$= (R_y - Q_z,\ P_z - R_x,\ Q_x - P_y)$$

Stokes' Theorem

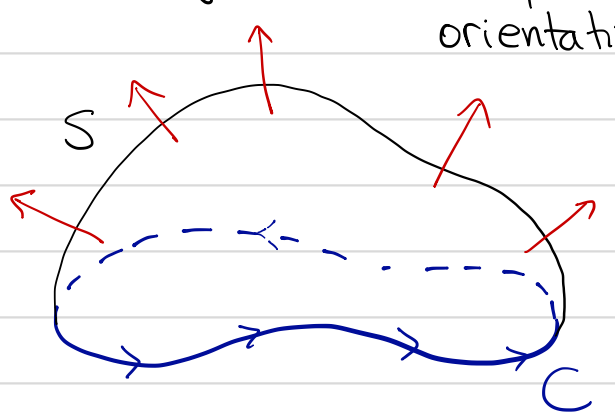
$$\boxed{\int_C \vec{F} \cdot d\vec{r} = \iint_S \operatorname{curl}(\vec{F}) \cdot d\vec{S}}$$

$\vec{F}$ vector field

S surface

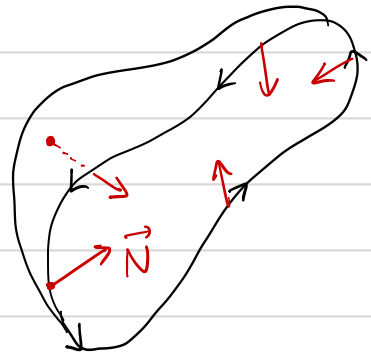
C boundary curve

} with compatible orientations

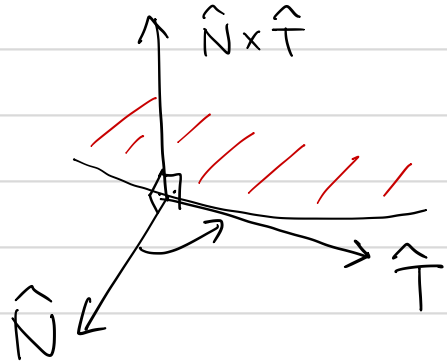


Orientation of C:

Walk along C, head in direction of $\vec{N}$ $\rightsquigarrow$ surface to your left.

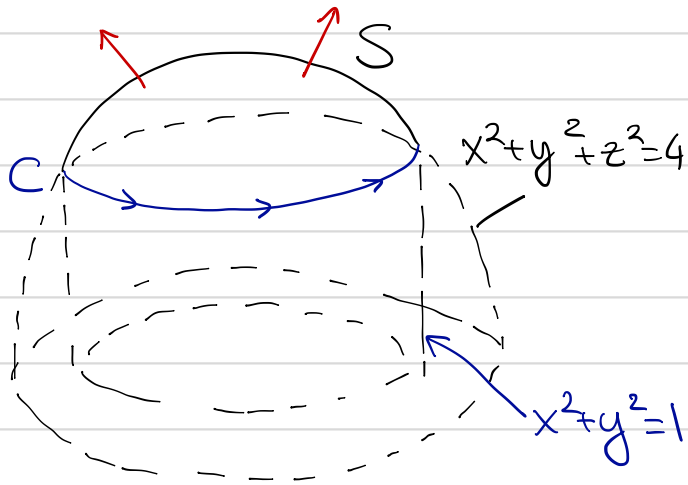


Right-hand rule:



$\hat{N} \times \hat{T}$ points <u>into</u> the surface.

Example: $\vec{F} = \langle xz, yz, xy \rangle$



$$\iint_S \nabla \times \vec{F} \cdot dS = \int_C \vec{F} \cdot d\vec{r} = 0$$

On C: $\underbrace{x^2+y^2}_{1} + z^2 = 4$

$\Rightarrow z = \sqrt{3}$ $\qquad dz = 0$

$x = \cos(t)$ $\qquad dx = -\sin(t)\,dt$

$y = \sin(t)$ $\qquad dy = \cos(t)\,dt$

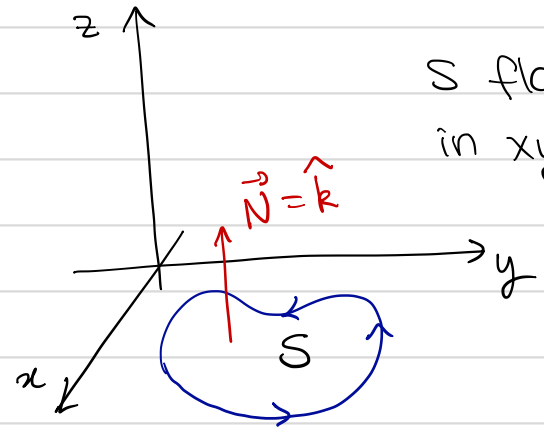
$$\int_C \vec{F} \cdot d\vec{r} = \int_C \langle xz, yz, xy \rangle \cdot \langle dx, dy, dz \rangle$$

$$= \int_0^{2\pi} \langle \sqrt{3}\cos t, \sqrt{3}\sin t, \sin t \cdot \cos t \rangle \cdot \langle -\sin t, \cos t, 0 \rangle \, dt$$

$$= \int_0^{2\pi} \left(\cancel{\sqrt{3}\cos t(-\sin t)} + \cancel{\sqrt{3}\sin t \cos t} + 0 \right) dt$$

$$= 0 \quad //$$

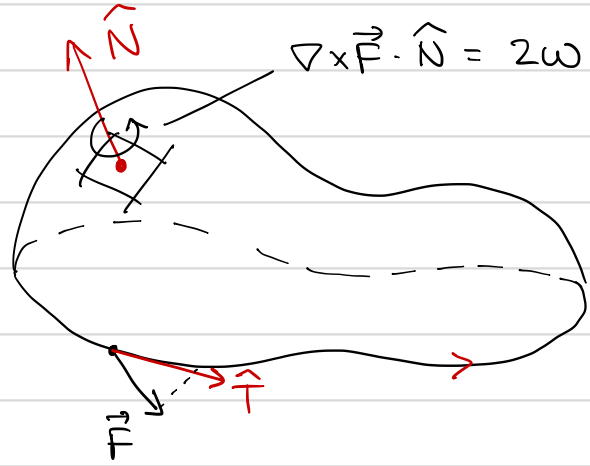
Relationship with Green's theorem



S flat, in xy-plane:

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \underbrace{(\nabla \times \vec{F}) \cdot \hat{k}}_{Q_x - P_y} \, dA$$

$\vec{F}$ = Fluid velocity



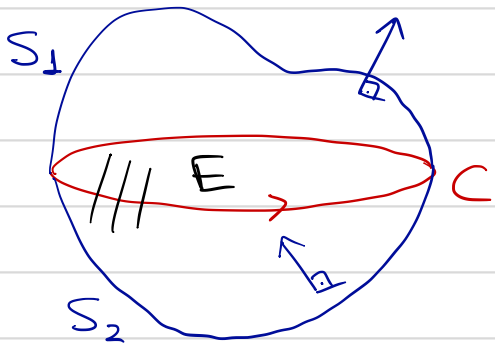
$$\iint_S \nabla \times \vec{F} \cdot \underbrace{\hat{N}\, dS}_{d\vec{S}} = \int_C \vec{F} \cdot \underbrace{\hat{T}\, ds}_{d\vec{r}}$$

$\nabla \times \vec{F} \cdot \hat{N} = 2\omega$

$= 2$(normal component of angular velocity)

$\vec{F} \cdot \hat{T}$ = tangential component of velocity.

Surface indepence in Stokes:



$$\underbrace{\iint_{S_1} \nabla\times\vec{F}\cdot\hat{N}\,dS}_{I_1} = \int_C \vec{F}\cdot d\vec{r}$$

$$= \underbrace{\iint_{S_2} (\nabla\times\vec{F})\cdot\hat{N}\,dS}_{I_2}$$

Another way to get that $I_1 = I_2$:

$$I_1 - I_2 = \iint_{S_1 - S_2} \nabla\times\vec{F}\cdot\hat{N}\,dS$$

$$= \iiint_E \underbrace{\nabla\cdot(\nabla\times\vec{F})}_{=0 \text{ for any } \vec{F}}\,dV = 0$$

Divergence theorem

E = solid region enclosed by S_1 & S_2.